Law of cosines

1 The law of sines

2

a $a^2 = h^2 + c^2 - 2cx + x^2$

b $h^2 = b^2 - x^2$

c $x = b\cos A$

d $a^2 = b^2 + c^2 - 2bc\cos A$

3 $\cos\theta = \dfrac{b^2 + c^2 - a^2}{2bc}$

4 $\cos U = -\dfrac{1}{15}$

5 $\angle ABC$

Unknown sides

6

a $a = 20.37$

b $a = 10.00$

c $a = 14.14$

d $a = 212.34$

e $q = 19.23$

f $c = 10.60$

g $a = 10.50$

h $y = 2.43$

7 $v = 5$

8 $x = 48.57$

9 $x = 45.7\text{ cm}$

Unknown angles

10 $\theta = 48.24^\circ$

11

a $w = 31.69^\circ$

b $A = 27.33^\circ$

c $w = 16.43^\circ$

d $B = 51.32^\circ$

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12 There is no solution because $\cos A = 1.1 > 1$. Alternatively, the two smaller sides must sum to more than the largest side to form a triangle.

13 $\theta = 24^\circ$

14 $x = 104^\circ$

15

a $x = 48^\circ$ **b** 132°

Applications

16 $x = 34.78$ km

17 $x = 40^\circ$

18 $x = 22^\circ$

19 $x = 37.79$ km

20

a $BD = 23.88$ m **b** $CD = 25.95$ m **c** 65.95 m

21 $x = 31$ mi

22

a $x = 264.9$ km **b** 586.0 km

23

a 3799 km **b** 4915.5 km **c** $x = 4014.7$ km

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